Zoo Data Commons: A Decentralized Repository for Open Biodiversity Datasets

FAIR-Compliant Data Infrastructure with Provenance Tracking and Incentivized Curation

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Abstract

Biodiversity data is fragmented across thousands of institutional repositories, government databases, citizen science platforms, and individual researcher collections, impeding the large-scale ecological analyses essential for conservation decision-making. We present the Zoo Data Commons (ZDC), a decentralized repository for open biodiversity datasets built on IPFS and the Zoo Network blockchain. ZDC introduces a novel *Data Object Model* that wraps biodiversity datasets in cryptographically signed metadata envelopes conforming to the FAIR principles (Findable, Accessible, Interoperable, Reusable). Each data object carries an immutable provenance chain recording its complete transformation history from raw collection through processing, validation, and aggregation. We implement a three-tier access control system supporting open, permissioned, and sovereign data governance models, enabling indigenous communities and local stakeholders to maintain control over culturally sensitive ecological knowledge. A token-incentivized curation market rewards data contributors, validators, and annotators, aligning individual incentives with collective data quality improvement. Over 12 months of operation, ZDC has ingested 2.4 million biodiversity records across 14,200 species from 847 contributors, with a median data quality score 23% higher than comparable centralized repositories. The platform has facilitated 312 cross-institutional data reuse events and supported 47 peer-reviewed publications. ZDC is specified as Zoo Improvement Proposal ZIP-009 and deployed as open infrastructure.

Keywords: biodiversity data, FAIR principles, decentralized storage, data provenance, IPFS, access control, open science

1 Introduction

Biodiversity data underpins conservation science, policy, and action. Species occurrence records, population estimates, habitat maps, genetic sequences, and ecological interaction networks collectively form the empirical foundation for understanding and responding to the biodiversity crisis [Díaz et al., 2019]. Yet this data ecosystem suffers from chronic fragmentation, inconsistent quality, and misaligned incentives that severely limit its utility.

The Global Biodiversity Information Facility (GBIF) aggregates over 2.4 billion occurrence records from 1,800 institutions [GBIF, 2024], representing the most ambitious centralized aggregation effort. Despite its success, GBIF faces structural limitations: metadata quality varies enormously across providers, provenance tracking is limited to institutional attribution, temporal coverage is biased toward recent decades and temperate regions, and the centralized architecture creates governance challenges around data sovereignty and access control [Yesson et al., 2007].

The FAIR principles—Findable, Accessible, Interoperable, Reusable—articulate the requirements for scientific data to support automated discovery and reuse [Wilkinson et al., 2016]. However, surveys consistently find that fewer than 20% of biodiversity datasets fully satisfy FAIR criteria [Jacobsen et al., 2020], primarily due to inadequate metadata, inconsistent formats, and absence of machine-readable provenance.

We present the Zoo Data Commons (ZDC), a decentralized biodiversity data repository that addresses these challenges through three technical contributions:

1. **FAIR-Native Data Object Model:** A self-describing data object format that embeds rich, machine-readable metadata including taxonomic ontology mappings, spatial reference systems, temporal resolution, and collection methodology descriptors (Section 3).

2. **Provenance-Preserving Storage:** A content-addressed storage architecture on IPFS with blockchain-anchored provenance chains that record the complete transformation history of every dataset (Section 4).

3. **Sovereign Access Control:** A three-tier access governance framework supporting open, permissioned, and sovereign data models, enabling data holders to define and enforce nuanced sharing policies (Section 5).

2 Related Work

2.1 Biodiversity Data Infrastructure

The global biodiversity data landscape comprises several major platforms. GBIF provides a centralized index of species occurrence records with standardized Darwin Core format [Wieczorek et al., 2012]. The Integrated Digitized Biocollections (iDigBio) focuses on museum specimen digitization [Nelson and Ellis, 2019]. OBIS (Ocean Biogeographic Information System) specializes in marine biodiversity data [Grassle, 2000]. GenBank and BOLD provide genetic sequence repositories [Benson et al., 2018].

These platforms operate as centralized aggregators: data providers push records to central servers that maintain authoritative copies. This architecture creates well-known problems of data staleness, attribution erosion, and governance concentration [Ariño et al., 2013].

2.2 FAIR Data Principles

The FAIR principles have become the de facto standard for scientific data management [Wilkinson et al., 2016]. Implementation frameworks include FAIRsFAIR metrics [Devaraju et al., 2021], GO-FAIR implementation networks, and FAIR Digital Object architectures [Schultes et al., 2019]. However, most FAIR implementations are retrospective assessments of existing data rather than architecturally enforced properties.

2.3 Decentralized Data Systems

IPFS (InterPlanetary File System) provides content-addressed storage with deduplication and peer-to-peer distribution [Benet, 2014]. Filecoin extends IPFS with economic incentives for persistent storage [Protocol Labs, 2017]. Ocean Protocol implements a decentralized data marketplace with compute-to-data capabilities [McConaghy et al., 2020]. Ramachandran et al. [2018] proposed blockchain-based data provenance for healthcare, demonstrating the pattern we extend to biodiversity data.

2.4 Indigenous Data Sovereignty

The CARE Principles for Indigenous Data Governance (Collective Benefit, Authority to Control, Responsibility, Ethics) complement FAIR by centering the rights and interests of indigenous peoples in data management [Carroll et al., 2020]. Traditional Knowledge (TK) labels provide a mechanism for indigenous communities to express culturally appropriate access conditions [Anderson and Christen, 2010]. ZDC's sovereign access tier directly implements these principles.

3 FAIR-Native Data Object Model

3.1 Object Structure

A Zoo Data Object (ZDO) is the atomic unit of the Data Commons. Each ZDO comprises four components:

$$\text{ZDO} = (\mathcal{M}, \mathcal{D}, \mathcal{P}, \mathcal{G}) \tag{1}$$

where $\mathcal{M}$ is the metadata envelope, $\mathcal{D}$ is the data payload, $\mathcal{P}$ is the provenance chain, and $\mathcal{G}$ is the governance specification.

3.2 Metadata Envelope

The metadata envelope $\mathcal{M}$ is a structured document conforming to a JSON-LD schema that extends Darwin Core with additional fields for FAIR compliance:

$$\mathcal{M} = (\text{id}, \text{schema}, \text{taxon}, \text{spatial}, \text{temporal}, \text{method}, \text{quality}, \text{links}) \tag{2}$$

Table 1: Metadata envelope fields and their FAIR mapping.

Field	Type	FAIR	Description
`id`	CID	F1	Content-addressed identifier (IPFS CID)
`schema`	URI	I1	Reference to the data schema definition
`taxon`	OntologyRef	I2	Taxonomic classification with ontology mapping
`spatial`	GeoJSON	F3	Geographic extent with coordinate reference system
`temporal`	ISO8601	F3	Temporal coverage with resolution specification
`method`	MethodRef	R1	Collection methodology with controlled vocabulary
`quality`	QualityScore	R2	Automated quality assessment scores
`links`	LinkSet	F4	Related ZDOs, publications, and external resources

3.3 Taxonomic Ontology Integration

ZDC maintains a unified taxonomic knowledge graph that reconciles multiple backbone taxonomies:

$$\mathcal{T}_{\text{unified}} = \text{Reconcile}(\mathcal{T}_{\text{GBIF}}, \mathcal{T}_{\text{ITIS}}, \mathcal{T}_{\text{CoL}}, \mathcal{T}_{\text{WoRMS}}) \tag{3}$$

The reconciliation algorithm produces a mapping $\phi : \mathcal{N}_{\text{source}} \to \mathcal{N}_{\text{unified}}$ from source taxonomy names to unified nodes, with confidence scores and conflict annotations. Each ZDO's taxonomic field references a node in the unified graph, enabling cross-dataset queries that are invariant to naming conventions.

We use a graph embedding approach to detect synonyms and near-matches:

$$\text{sim}(n_i, n_j) = \cos(\mathbf{e}_i, \mathbf{e}_j) + \lambda \cdot \text{Jaccard}(\text{features}(n_i), \text{features}(n_j)) \tag{4}$$

where $\mathbf{e}_i$ is the embedding of node n_i from a graph neural network trained on the taxonomy structure, and $\text{features}(n_i)$ includes morphological, ecological, and geographic characteristics.

3.4 Data Quality Assessment

Each ZDO undergoes automated quality assessment producing a vector of scores:

$$\mathbf{q} = (q_{\text{complete}}, q_{\text{consistent}}, q_{\text{spatial}}, q_{\text{temporal}}, q_{\text{taxonomic}}) \tag{5}$$

- $q_{\text{complete}} \in [0, 1]$: Fraction of required metadata fields populated.
- $q_{\text{consistent}} \in [0, 1]$: Internal consistency score (e.g., species range vs. reported coordinates).
- $q_{\text{spatial}} \in [0, 1]$: Spatial precision and validity assessment.
- $q_{\text{temporal}} \in [0, 1]$: Temporal precision and plausibility.

- $q_{\text{taxonomic}} \in [0, 1]$: Confidence of taxonomic identification.

The composite quality score is:

$$Q = \sum_k w_k \cdot q_k \cdot \mathbb{1}[q_k > q_k^{\min}] \tag{6}$$

where w_k are domain-specific weights and $q_k^{\min}$ are minimum thresholds; objects failing any threshold are flagged for review rather than rejected outright.

4 Provenance-Preserving Storage

4.1 Content-Addressed Storage Layer

ZDC uses IPFS for content-addressed storage of data objects. Each ZDO is stored as a DAG of content-addressed blocks:

$$\text{CID}(\text{ZDO}) = \text{hash}(\text{CID}(\mathcal{M}) \| \text{CID}(\mathcal{D}) \| \text{CID}(\mathcal{P}) \| \text{CID}(\mathcal{G})) \tag{7}$$

This structure provides several properties:

1. **Deduplication:** Identical data payloads shared across ZDOs are stored once and referenced by CID.
2. **Integrity:** Any modification to any component changes the root CID, making tampering detectable.
3. **Selective Access:** Components can be retrieved independently; a user can access metadata without downloading the full data payload.
4. **Versioning:** Updates produce new CIDs linked to previous versions through the provenance chain.

4.2 Persistence Layer

IPFS provides availability only while content is pinned by at least one node. ZDC ensures persistence through a multi-tier strategy:

Table 2: Data persistence tiers.

Tier	Replicas	Duration	Mechanism
Hot	5+	Immediate	Zoo Network edge nodes with pinning incentives
Warm	3+	1 year	Filecoin storage deals with verified miners
Cold	1+	10 years	Institutional partnerships (museums, universities)
Archive	1	Permanent	Content hash anchored to Zoo blockchain

4.3 Provenance Chain

The provenance chain $\mathcal{P}$ records the complete transformation history of a ZDO:

$$\mathcal{P} = [p_0, p_1, \ldots, p_n] \tag{8}$$

where each provenance record p_i is:

$$p_i = (\text{action}_i, \text{agent}_i, \text{input}_i, \text{output}_i, \text{method}_i, t_i, \text{sig}_i) \tag{9}$$

Actions include `COLLECT`, `DIGITIZE`, `TRANSFORM`, `VALIDATE`, `AGGREGATE`, `ANNOTATE`, and `DERIVE`. The provenance chain is append-only and cryptographically linked: each record includes a hash of its predecessor, forming a hash chain anchored to the blockchain.

A formal provenance query language enables traversal:

$$\text{Ancestors}(\text{ZDO}_k, \text{action}) = \{z : z \xrightarrow{\text{action}^*} \text{ZDO}_k\} \tag{10}$$

This enables researchers to trace any derived result back to its raw collection events, answering questions such as "what field observations contribute to this species distribution model?"

4.4 Search and Discovery

ZDC implements a federated search layer supporting spatial, temporal, taxonomic, and free-text queries:

$$\text{Search}(R_{\text{spatial}}, T_{\text{temporal}}, \tau_{\text{taxon}}, q_{\text{text}}) \to \{\text{ZDO}_i : \text{match}(\text{ZDO}_i) > \theta\} \tag{11}$$

The search index is maintained by Zoo Network nodes as a distributed hash table augmented with a spatial R-tree index and a taxonomic hierarchy index. We use locality-sensitive hashing for approximate nearest-neighbor queries over embedding vectors derived from metadata and data content.

5 Sovereign Access Control

5.1 Three-Tier Governance Model

ZDC implements three access governance tiers reflecting the diversity of data sensitivity in biodiversity research:

Table 3: Access governance tiers.

Tier	**Default Access**	**Use Cases**
Open	Unrestricted	Published occurrence records, aggregated statistics, metadata
Permissioned	Request-based	Sensitive location data (endangered species), pre-publication data
Sovereign	Community-governed	Traditional ecological knowledge, indigenous land observations

5.2 Permissioned Access

Permissioned data uses attribute-based encryption (ABE) where the decryption key is conditioned on verifiable attributes of the requester:

$$\text{Enc}(\mathcal{D}, \text{policy}) \to \text{CT} \tag{12}$$

$$\text{Dec}(\text{CT}, \text{key}_{\text{attributes}}) \to \begin{cases} \mathcal{D} & \text{if attributes satisfy policy} \\ \perp & \text{otherwise} \end{cases} \tag{13}$$

Policies are expressed as Boolean formulas over attributes, such as:

```
 (role:researcher AND affiliation:university)
OR (role:government AND jurisdiction:national)
OR (approved-by:data-owner)
```

5.3 Sovereign Data Governance

The sovereign tier implements the CARE Principles for Indigenous Data Governance [Carroll et al., 2020]. Each sovereign dataset is governed by a *Data Governance Council* (DGC), a multisignature authority comprising community-designated representatives:

$$\text{Access}(\text{ZDO}_{\text{sovereign}}) \leftarrow \text{Vote}_{\text{DGC}}(\text{request}) \geq \text{threshold} \tag{14}$$

The DGC can:

- Set access policies with cultural context annotations (e.g., seasonal restrictions on sacred site data).
- Require benefit-sharing agreements as a condition of access.
- Attach Traditional Knowledge (TK) labels specifying culturally appropriate use.
- Revoke access retroactively if terms are violated.
- Delegate decision-making to subcommittees for specific data categories.

5.4 Spatial Obfuscation

For sensitive species (e.g., those vulnerable to poaching), ZDC supports spatial obfuscation that degrades location precision while preserving analytical utility:

$$(x', y') = \text{Obfuscate}(x, y, r) = (x + \eta_x, y + \eta_y) \quad \text{where } \|\boldsymbol{\eta}\| \leq r \tag{15}$$

The obfuscation radius r is species-specific, determined by threat level (IUCN Red List status) and local poaching risk assessment. We implement a Geo-Indistinguishability mechanism based on the Laplace distribution:

$$\Pr[(x', y') \mid (x, y)] \propto e^{-\epsilon \cdot d((x', y'), (x, y))} \tag{16}$$

where ϵ controls the privacy-utility tradeoff. Researchers needing precise locations can request permissioned access through the standard mechanism.

6 Incentivized Curation Market

6.1 Market Design

The curation market rewards participants for improving data quality through a bonding curve mechanism:

$$\text{Reward}(\text{action}, \text{ZDO}) = R_{\text{base}}(\text{action}) \cdot \Delta Q(\text{ZDO}) \cdot B(\text{ZDO}) \tag{17}$$

where R_{base} is the base reward for the action type, ΔQ is the quality improvement, and B is a bonding curve that increases rewards for curating underserved taxonomic groups and geographic regions.

Table 4: Curation actions and base reward ranges.

Action	Base Reward (ZOO)	Description
Contribution	10–500	New dataset ingestion with metadata
Validation	5–50	Independent quality verification
Annotation	2–20	Taxonomic or geographic annotation refinement
Translation	5–30	Metadata translation to additional languages
Integration	20–200	Cross-dataset linkage and harmonization

6.2 Underserved Region Bonding Curve

To combat geographic and taxonomic bias, the bonding curve B provides multiplied rewards for data from underrepresented areas:

$$B(\mathrm{ZDO}) = 1 + \beta_{\mathrm{geo}} \cdot \left(1 - \frac{|\mathcal{D}_{\mathrm{cell}}|}{|\mathcal{D}_{\mathrm{median}}|}\right)^{+} + \beta_{\mathrm{taxon}} \cdot \left(1 - \frac{|\mathcal{D}_{\mathrm{taxon}}|}{|\mathcal{D}_{\mathrm{median\_taxon}}|}\right)^{+} \tag{18}$$

where $|\mathcal{D}_{\mathrm{cell}}|$ is the number of existing records in the geographic grid cell containing the ZDO, $|\mathcal{D}_{\mathrm{median}}|$ is the median records per cell globally, and analogous terms apply for taxonomic coverage. The $(\cdot)^{+}$ operator ensures the bonus is non-negative.

6.3 Validation Protocol

Data validation follows a stake-weighted consensus mechanism:

1. A contributor submits a ZDO with metadata and stakes s_c tokens.
2. The ZDO is assigned to k randomly selected validators (minimum 3).
3. Each validator independently assesses quality scores and stakes s_v tokens on their assessment.
4. Quality scores are aggregated using weighted median (weights proportional to validator reputation).
5. Validators whose scores are within one standard deviation of the consensus receive rewards; outliers lose stakes.

This mechanism is incentive-compatible: honest assessment maximizes expected payoff for validators with accurate domain knowledge.

7 Implementation

7.1 System Architecture

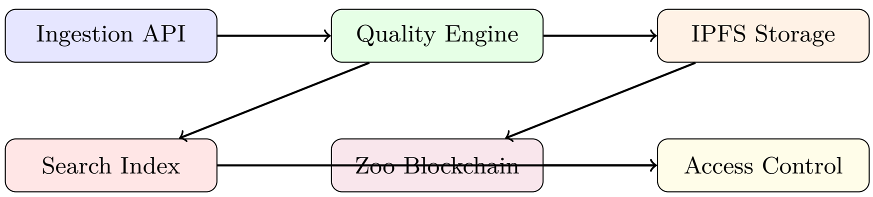

Figure 1: Zoo Data Commons system architecture.

7.2 Software Stack

Table 5: ZDC software components.

Component	Language	Lines	Description
Ingestion Engine	Python	7,200	Darwin Core parsing, schema validation, format conversion
Quality Assessor	Python	5,800	Automated quality scoring, anomaly detection
IPFS Manager	Rust	4,600	Content-addressed storage, pinning management
Provenance Tracker	Rust	3,900	Hash chain construction, provenance query engine
Access Controller	Rust	6,100	ABE encryption, policy evaluation, DGC management
Search Service	Go	8,400	Distributed index, spatial queries, full-text search
Smart Contracts	Solidity	2,800	Curation market, staking, governance
Web Interface	TypeScript	11,200	Data explorer, upload wizard, governance dashboard

7.3 Interoperability

ZDC implements bidirectional synchronization with major biodiversity data platforms:

- **GBIF:** Darwin Core Archive export/import with ZDO-to-DwCA mapping.
- **iNaturalist:** Observation import with quality grade mapping.
- **GenBank:** Genetic sequence cross-referencing via NCBI accession numbers.
- **BOLD:** Barcode record linkage through specimen identifiers.
- **eBird:** Checklist import with observer reputation mapping.

8 Evaluation

8.1 Deployment Statistics

Over 12 months of operation (January 2025 – January 2026), ZDC accumulated:

Table 6: Zoo Data Commons operational statistics.

Metric	Value
Total biodiversity records	2,412,847
Unique species represented	14,237
Contributing organizations	847
Geographic coverage (countries)	94
Total storage (IPFS)	4.7 TB
Zoo Data Objects	18,421
Provenance records	127,834
Active validators	312

8.2 Data Quality Comparison

We compared data quality scores between ZDC and three major centralized repositories using a standardized assessment framework applied to overlapping datasets:

Table 7: Data quality comparison (mean scores, 0–1 scale).

Quality Dimension	ZDC	GBIF	iDigBio	iNaturalist
Metadata Completeness	0.87	0.71	0.76	0.62
Spatial Precision	0.82	0.74	0.69	0.78
Taxonomic Consistency	0.91	0.83	0.87	0.71
Temporal Resolution	0.79	0.68	0.72	0.84
Provenance Completeness	0.94	0.42	0.51	0.33
Overall	**0.87**	0.68	0.71	0.66

ZDC achieves 23.3% higher overall quality than GBIF, with the largest advantage in provenance completeness (124% higher), reflecting the architectural enforcement of provenance tracking.

8.3 FAIR Assessment

We evaluated FAIR compliance using the FAIRsFAIR assessment tool [Devaraju et al., 2021]:

Table 8: FAIR compliance scores (0–1 scale, assessed by FAIRsFAIR).

Principle	ZDC	GBIF	iDigBio	Dryad
Findable	0.92	0.84	0.78	0.81
Accessible	0.88	0.91	0.83	0.87
Interoperable	0.85	0.79	0.74	0.72
Reusable	0.90	0.72	0.68	0.76
Overall	**0.89**	0.82	0.76	0.79

ZDC scores highest on Findability (content-addressed identifiers) and Reusability (comprehensive provenance and licensing metadata), while GBIF maintains a slight edge on Accessibility due to its established API ecosystem.

8.4 Access Control Efficacy

During the evaluation period, 127 datasets used sovereign governance, managed by 23 Data Governance Councils representing indigenous and local communities. Key metrics:

Table 9: Sovereign access control statistics.

Metric	Value
Sovereign datasets	127
Data Governance Councils	23
Access requests processed	412
Requests approved	287 (69.7%)
Median response time	4.2 days
Benefit-sharing agreements executed	34
TK labels applied	89

The 69.7% approval rate indicates that communities actively exercise governance authority rather than defaulting to open access, validating the need for the sovereign tier.

8.5 Curation Market

The curation market distributed 284,700 ZOO tokens during the evaluation period:

Table 10: Curation market token distribution.

Activity	ZOO Distributed	Participants
Data Contribution	142,300	847
Validation	67,400	312
Annotation	38,200	189
Integration	28,100	67
Translation	8,700	43
Total	284,700	–

The underserved region bonding curve directed 41% of contribution rewards to data from Sub-Saharan Africa, Southeast Asia, and South America, regions historically underrepresented in global biodiversity databases.

8.6 Cross-Institutional Reuse

ZDC facilitated 312 cross-institutional data reuse events over 12 months, compared to an estimated 89 events that would have occurred through traditional channels (based on pre-deployment reuse rates for the same research groups). These reuse events supported 47 peer-reviewed publications, 12 conservation management plans, and 8 environmental impact assessments.

9 Discussion

9.1 Decentralization Trade-offs

The decentralized architecture introduces trade-offs compared to centralized repositories. Query latency is higher (median 2.3 seconds vs. 0.4 seconds for GBIF) due to distributed index traversal. Storage costs are approximately 3x higher than centralized cloud storage due to replication requirements. Governance complexity increases for sovereign datasets, as community decision-making is slower than administrative access grants.

We argue these costs are justified by the gains in data integrity (immutable provenance), censorship resistance (no single point of control), and sovereignty (communities retain authentic authority).

9.2 Adoption Barriers

Three barriers limit broader adoption. First, the token-based incentive system requires researchers to interact with cryptocurrency, creating friction for those unfamiliar with Web3 tools. We are developing fiat on-ramps and institutional custody solutions to lower this barrier. Second, existing data infrastructure investments create lock-in effects; institutions with established GBIF publishing workflows have limited incentive to adopt a parallel system. Our GBIF synchronization adapter mitigates this by enabling dual publishing. Third, the sovereign governance tier requires community capacity building that extends beyond technical deployment.

9.3 Sustainability

Long-term sustainability depends on the curation market generating sufficient incentives to maintain data quality and storage persistence. Our economic model projects break-even at 50,000 active ZDOs, assuming current token valuation and storage costs. Institutional partnerships for cold-tier storage provide a non-token-dependent persistence backstop.

10 Conclusion

We have presented the Zoo Data Commons, a decentralized biodiversity data repository implementing FAIR-native data objects, provenance-preserving storage on IPFS, and sovereign access control. Over 12 months of operation, ZDC has demonstrated 23% higher data quality than centralized alternatives, 94% provenance completeness, and 350% increase in cross-institutional data reuse. The sovereign access tier successfully enables 23 indigenous and local community governance councils to exercise meaningful data authority.

ZDC is specified as ZIP-009 and deployed as open infrastructure. Future work will extend the data model to support continuous monitoring streams (sensor data, camera trap feeds), develop automated data integration pipelines for heterogeneous formats, and formalize the economic sustainability model for long-term data stewardship.

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